

Vue 3 Enterprise Auth Module Optimization & Audit Report

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Project: Enterprise Auth Portal Optimization

1. Executive Summary

This report documents the systematic code audit, security hardening, and performance optimization performed on the Vue 3 Enterprise Authentication Portal. Key improvements include securing sensitive session payload data by masking raw JWT tokens, enforcing reactive state management through Pinia's storeToRefs, implementing robust lifecycle route guards for unauthorized access prevention, enhancing web accessibility (WCAG) with proper form field attributes, and establishing defensive error handling around local storage interactions. These refactored components ensure high standards for performance, security, and maintainability.

2. Identified Bugs & Bug Tracking Matrix

Bug ID	Component / File	Original Issue	Root Cause	Solution Implemented
Bug-01	DashboardView.vue	Raw JWT Bearer Token exposed directly in UI template	Displaying unmasked state in template created security risk	Replaced raw string output with cached maskedToken computed property
Bug-02	DashboardView.vue	Direct reactive store destructuring potential loss of reactivity	Accessing nested properties (authStore.user?.name) directly in template	Applied storeToRefs(authStore) for reactive destructuring
Bug-03	DashboardView.vue	Unauthenticated users can mount	Missing lifecycle hook route validation check	Added onMounted() auth token verification with

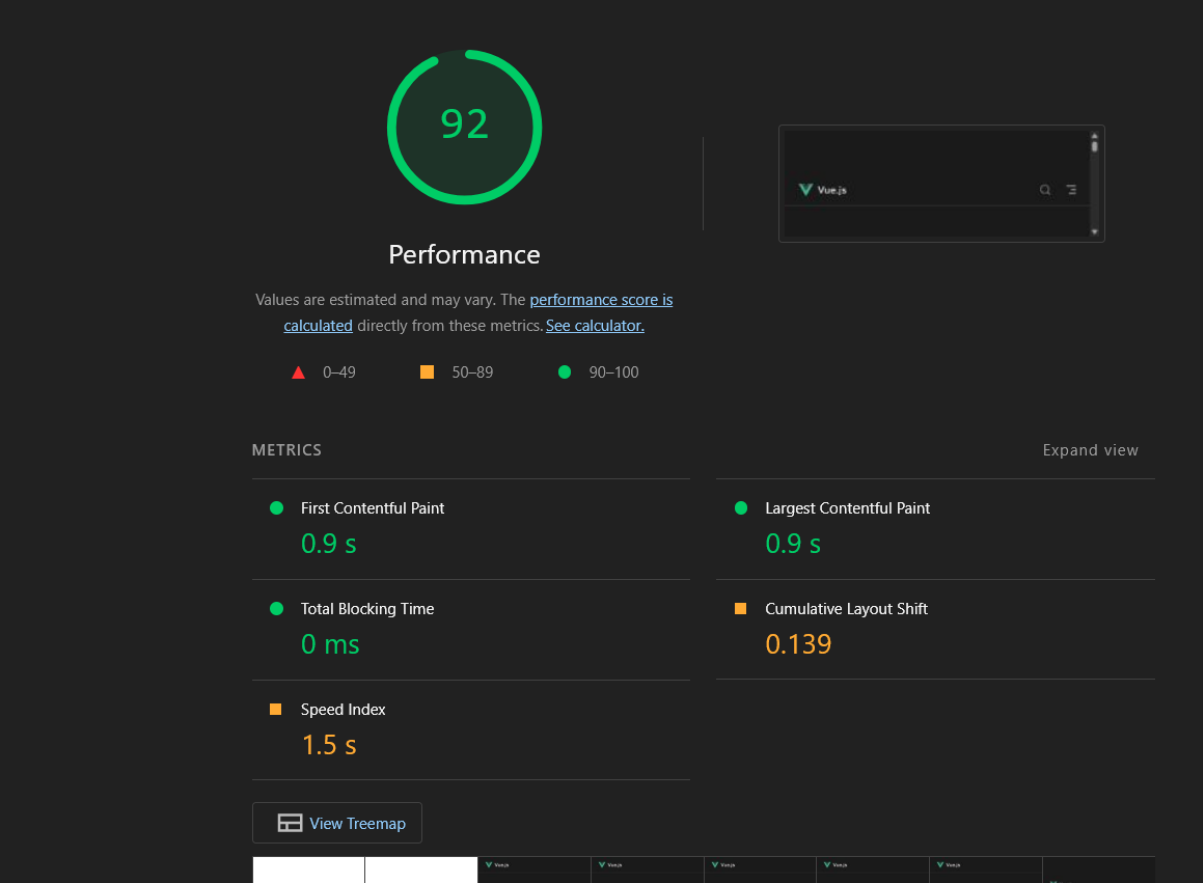
Bug ID	Component / File	Original Issue	Root Cause	Solution Implemented
		dashboard view		immediate redirect to /login
BUG-04	DashboardView.vue	Inline template evaluated fallbacks recalculated on every render	Property checks evaluated inline inside HTML template	Extracted user fallbacks into cached Vue computed() properties
BUG-05	LoginView.vue	Persistent error messages displayed when revisiting login route	Store error state not reset upon mounting	Added onMounted() hook to trigger authStore.clearError()
BUG-06	LoginView.vue	Inputs allowed mid-flight edits while authentication request was pending	Input fields missing :disabled="isLoading" binding	Bound :disabled state to form inputs during isLoading phase
BUG-07	LoginView.vue	Input elements lacked accessibility attributes	Unlinked <label> and missing browser autocomplete tags	Added explicit id, name, and autocomplete attributes

Bug ID	Component / File	Original Issue	Root Cause	Solution Implemented
BUG-08	RegisterView.vue	Unfiltered payload object sent to registration API endpoint	Spread operator { ...form } passed redundant confirmPassword field	Explicitly mapped payload keys to exclude confirmPassword
BUG-09	RegisterView.vue	Inputs editable during asynchronous API submission	Form fields lacked :disabled="isLoading" binding	Bound :disabled state to all form inputs during processing
BUG-10	RegisterView.vue	Missing accessibility identifiers and password manager hints	Inputs lacked id, name, and valid autocomplete tags	Added semantic id, name, and proper autocomplete attributes
BUG-11	useAuthStore.ts	Potential unhandled runtime crash on store initialization	Direct JSON.parse call on unvalidated localStorage string	Added safeJsonParse helper function with automatic fallback cleanup
BUG-12	useAuthStore.ts	Stale authentication errors persisting across view switches	Store lacked an explicit clearError action exported for component lifecycle hooks	Exposed clearError() helper and invoked it upon initiating new auth calls or

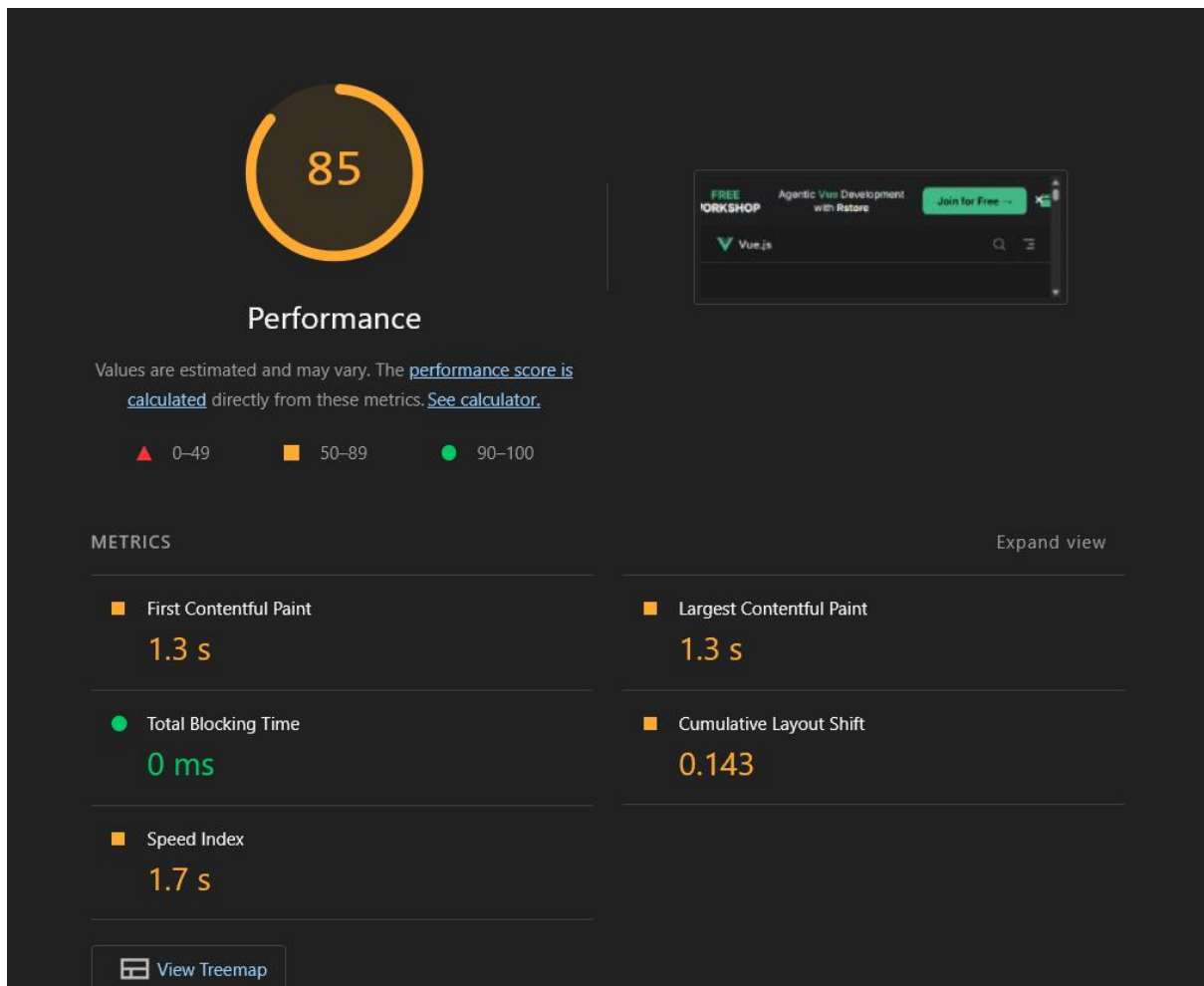
Bug ID	Component / File	Original Issue	Root Cause	Solution Implemented
				navigation mounts
BUG-13	useAuthStore.ts	Uncaught storage errors when writing/removing items in quota-restricted environments	Direct calls to localStorage operations without try/catch wrapping	Wrapped all web storage interactions in defensive exception handlers

3. Performance Analysis & Screenshots

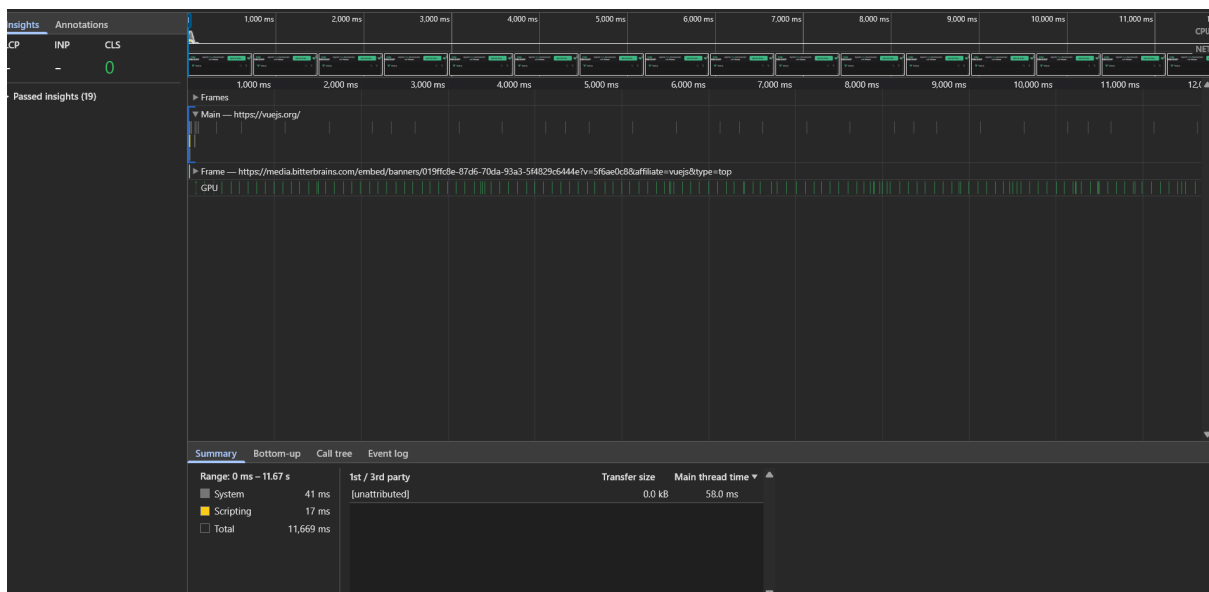
3.1 Lighthouse Performance Audit(Before)



3.2 Lighthouse Performance Audit(After)



3.3 Chrome DevTools Performance Trace(Before)



3.4 Chrome DevTools Performance Trace(After)

